

Job Profile

Valid from: - REVIEWED -

Basic data	Name of Function	Software Architect (SWA)	
	Function Area	Reseach / Development	
	Operational Field	all	
	Function Type	Expert / Senior Expert	
	Operational Area	National & International	
	Business Type	all	
	Objectives of the function (Mission)	Software Architect (SWA) The Software Architect drives and guides the specification and implementation of a software architecture for a project The Software Architect provides the architecture vision of a whole project or a subsystem over all its domain and technical aspects and ensure this vision's sustainability His/her decisions are driven by a clear focus on the intended business with the software and its associated requirements comming from product managers, product lifecycle managers. He/she is guided by maturity, vision, depth of domain and technical experience and the ability to identify, analyze, and decide on relevant issues in time and with courage, even in the absense of complete information. The Software Architect coaches, and motivates developers in his/her team over the entire lifetime of a project to enable, support, and enforce an appropriate implementation, maintenance, quality assurance, test, and evolution of the software architectures he/she designed, even in the face of challenging, changing, and evolving business cases, requirements, realization technologies Before taking important decisions, that are costly to change, he/she involves the project manager in the decision making process.	
	Indicators (Project category A: - order value : very high - risk : very high - complexity : very high)	Project Category: class B and class C projects; Project Scope: business unit; Project Type: product development, release development	
	Contacts (internal /external)	Requirements Engineer, Project Manager,Developer,Testmanager, Product Manager, System Architect	
	Responsibility Area / Tasks		
	Software Architect (SWA)	level of responsibility	
Business understanding the business cases & domain	- Validation of technology roadmaps	responsible	
	- Implementation of the IPR and business protection strategy	responsible	
	- Evaluation of selected technologies (e.g. by prototypes)	responsible	
	- Supplier evaluation - Make-or-Buy decision	involved involved	
Requirements Engineering understanding and clarifying requirements	- Definition of system architecture	involved	
	- Derive software requirements and SW functional specification	responsible	
	- Define external SW interfaces	responsible	
	- Validate SW requirements	responsible	
	- Prove SW feasibility	responsible	
Design Designing a concrete software architecture / supervising ist design	- System lifecycle planning (field installation, maintenance, sun setting)	involved	
	- Patent identification and creation	responsible	
	- Determine SW architecture and design	responsible	
	- Create design prototypes	responsible	
	- Define architecture and design rationale	responsible	
	- Establish architecture traceability	responsible	
	- Identification and specification of interfaces	responsible	
- Allocation of requirements to increments	responsible		
- Specification of system architecture & interfaces	involved		
	- Establish integration sequence	responsible	
	- Provide detailed planning information for tracking of integration by Project Management	responsible	

		- Setup of a supplier agreement	involved
	Implement updating SW concept	- Review detailed SW design - Implement critical algorithms/components - Setup SW integration plan - Control SW integration - Update of increment requirements - Code reviews, resolution of unit- and integration test defects, SW integration, system integration, integration test specification - Validate deliveries of external suppliers - Technical coaching of implementation team	responsible involved responsible responsible responsible involved responsible responsible
	Test & Quality planning and supervising the integration of the implemented SW artefacts to a coherent system	- Monitoring of system integration - Driving the architecture and interface quality - Updating of SW deliveries for integration according to defects resolved - Integration test, resolution of integration test defects, system integration test	responsible responsible responsible involved
	Maintenance avoiding architecture drift	- Continuous architecture management, monitoring of internal software quality and structural changes	responsible
Competencies			
Knowledge		Software Architect (SWA) (to be enhanced by business unit specific needs, e.g. domain knowledge)	Level B/A/E
	Professional Knowledge	Business Case Understanding (e.g. marketing, business planning, competitor analysis)	basic
	Professional Knowledge	Global development (e.g. Outsourcing / Offshoring, Collaboration)	basic
	Professional Knowledge	Legal issues (Supplier agreements, license issues, claim management)	basic
	Professional Knowledge	Product Management (e.g. product lifecycle management)	basic
	Professional Knowledge	Requirements Engineering and use cases	advanced
	Professional Knowledge	Architecture and Design Competencies	advanced
	Professional Knowledge	Software development processes (e.g. SW lifecycle models, CMMi)	advanced
	Professional Knowledge	Test processes and methods	advanced
	Professional Knowledge	Product Line Engineering	basic
	Professional Knowledge	Communication skills	advanced
	Technologies	Development processes, covering the system-, electrical and mechanical disciplines	basic
	Methods	SW Design Methods (e.g. OOA/OOD/UML, Design Patterns)	expert
	Methods	Configuration and Change Management	basic
	Methods	Project Management (Methods and Tools)	basic
	Methods	Quality Assurance	basic
	Experience	Professional experience	Acquisition of technical knowledge in 4-5 years of professional experience in processing, installation, engineering and development in different roles and functions.
Project experience		Has minimum 2 years experience in successfully working as an architect for a project of category C.	
Leadership experience		Has successfully worked as a technical team lead team at least for 2 years (development teams).	
Intercultural experience		Has worked and lived abroad: optional Has cooperated in international teams: Recommended	
Which of the following capabilities are the most relevant for this function ? (Please select maximum 6 out of 18)			
Capabilities	Energy		
	Execute	Analytics Decision Making: Decision Enforcement	
	Energize	Coaching and Mentoring Communication Skills	
	Passion		
	Edge		
Education Requirements			
Education			
Additional Requirements			
Additions			